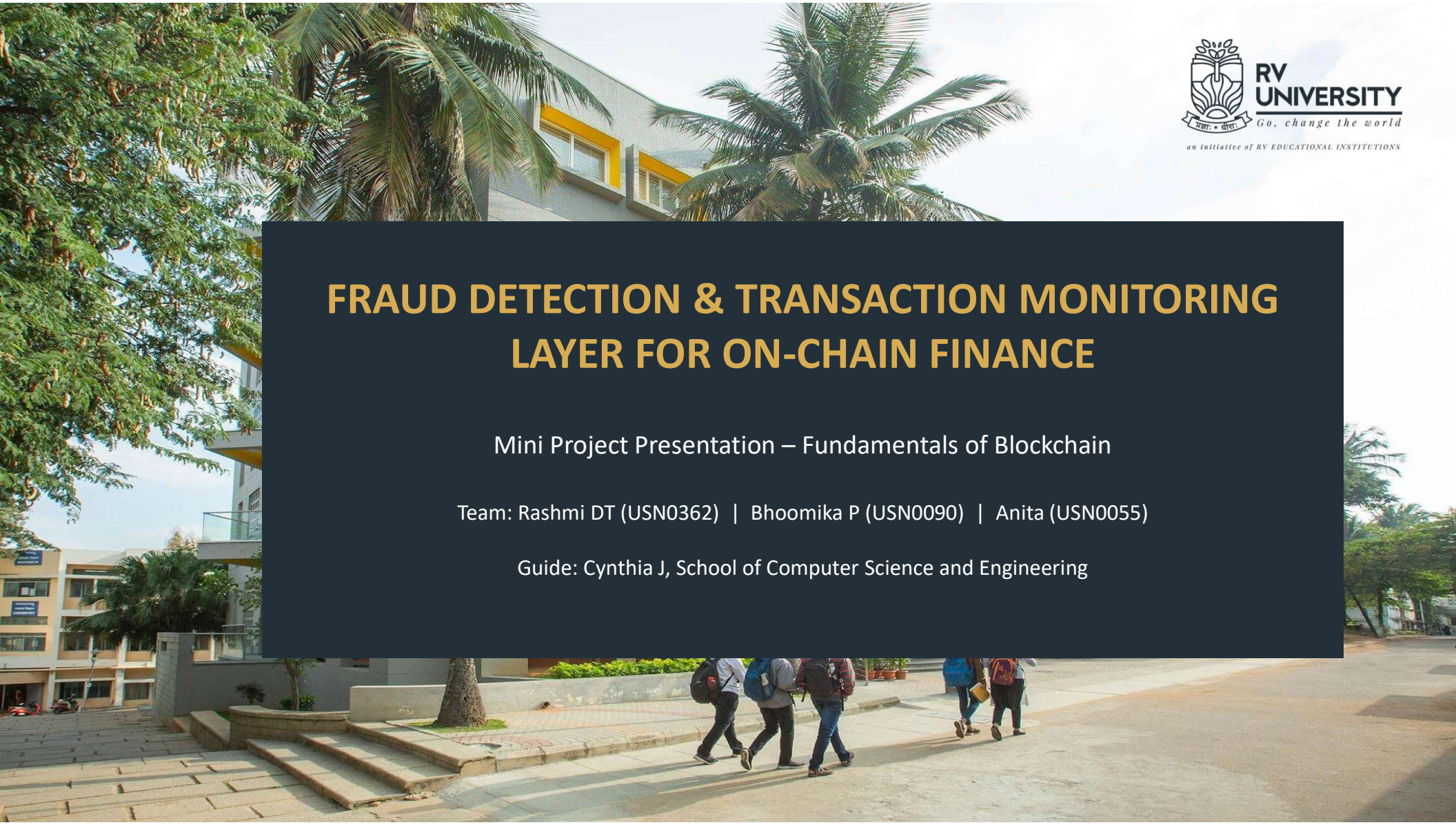


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FRAUD DETECTION & TRANSACTION MONITORING LAYER FOR ON-CHAIN FINANCE

Mini Project Presentation – Fundamentals of Blockchain

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Problem Statement

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PROBLEM STATEMENT

Ethereum is public and permissionless: anyone can create a new wallet and split or move funds in seconds, and there is no central compliance checkpoint. Exchanges and compliance analysts must find laundering patterns in raw transaction data by hand or with opaque tools, and nothing guarantees that a flagged alert or its review decision was not edited later.

Who is affected?

- Compliance officers at exchanges and wallet providers (VASPs)
- DeFi protocols and their users exposed to illicit flows
- Auditors and regulators who need evidence of what was flagged and who decided

Current pain points

- Manual review of public chain data is slow
- Rule-only or black-box scores are hard to justify to a regulator
- Alert logs live in editable databases
- Exchanges and bots trigger false alarms

Gap to address

- Missing today: explainable, real-time scoring with a tamper-evident review trail in one workflow
- This project: live mainnet monitoring, a 0–100 score with named reasons, an analyst case workflow, and alerts anchored in a smart contract

Problem Linkage

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FinTech Link	Anti-money-laundering (AML) transaction monitoring for crypto payments and DeFi. The stakeholder is the compliance analyst at an exchange or custodian, who must detect, justify and document suspicious activity (FATF risk-based approach for virtual assets).
Blockchain Link	Immutability and transparency of the audit trail. The RiskRegistry contract stores only a transaction hash, the flagged wallet, a 0–100 score, a reason bitmask and a review status. Role-based access (owner / analyst), duplicate protection, final review states, and events make alerts and decisions impossible to silently alter or delete.
External Data Link (Live / Off-chain)	Source: Ethereum mainnet blocks via Alchemy JSON-RPC (read-only, https). Live data, polled about every 12 s. Detection runs off-chain in Python. Alerts reach the contract through an analyst-approved queue (outbox.json) signed only by the anchor script on Sepolia testnet; no mainnet keys or funds are involved.

System Architecture

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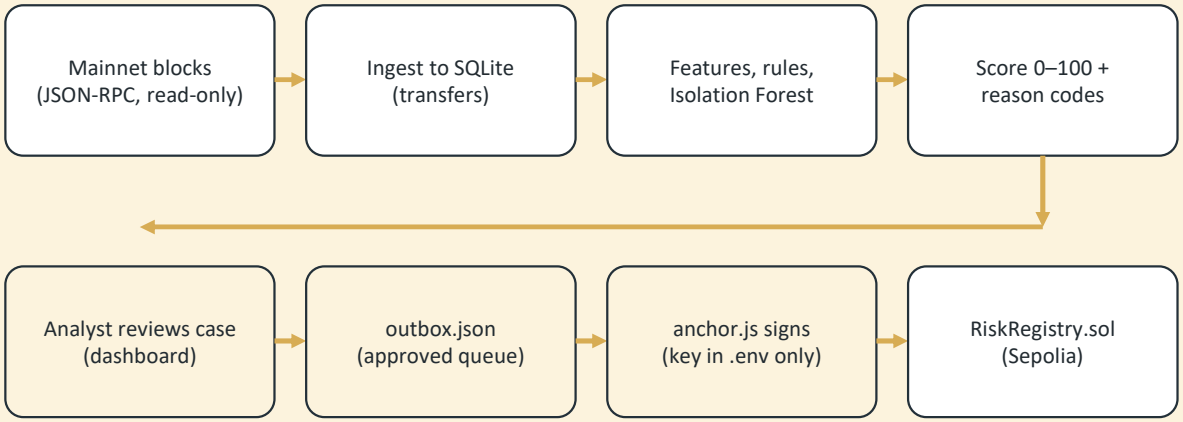
Data source
Ethereum mainnet (Alchemy RPC)

Ingestion
Python + SQLite

Detection
Rules + Isolation Forest

Analyst dashboard
Cases + review

Smart contract
RiskRegistry (Sepolia)



The web server holds no private key: only anchor.js signs transactions.
The proof (alert ID, tx hash, gas) flows back via anchored.json to the dashboard.

Component Description

- Ingestion reads real mainnet blocks over read-only RPC and stores transfers.
- Detection applies 8 rules and an Isolation Forest; a noisy-OR formula turns signals into one explained score.
- Dashboard groups alerts into one case per wallet and mirrors the contract's review rules.
- RiskRegistry stores hash, wallet, score, reason bitmask and status; emits events for every change.

Smart Contract Design

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Component	Type	Purpose
`owner / isAnalyst`	State variable	Roles: owner manages analysts and watchlist; analysts write alerts
`_alerts, _alertIdPlusOne`	State variable	Stores each alert; maps tx hash to id so a transaction is reported once
`raiseAlert()`	Function	Validates hash, wallet, score 1–100, non-zero reasons; rejects duplicates; stores alert
`updateStatus()`	Function	Open→Reviewed/Escalated/Dismissed; Escalated and Dismissed are final
`AlertRaised / AlertStatusChanged`	Event	Emitted on every alert and every review decision (audit trail)
`onlyAnalyst / onlyOwner`	Modifier	Only approved analysts write; only the owner edits roles and watchlist

Key Solidity Snippet

```
function raiseAlert(bytes32 txHash,
  address subject, uint8 riskScore,
  uint16 reasonMask)
  external onlyAnalyst returns (uint256 id)
{
  if (riskScore == 0 || riskScore > 100)
    revert InvalidRiskScore(riskScore);
  if (reasonMask == 0) revert NoReasonGiven();
  uint256 e = _alertIdPlusOne[txHash];
  if (e != 0) revert AlreadyReported(txHash, e-1);
  id = _alerts.length;
  _alerts.push(Alert(txHash, subject, ...));
  _alertIdPlusOne[txHash] = id + 1;
  emit AlertRaised(id, txHash, subject,
                   riskScore, reasonMask);
}
```

Testing & Results

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TC ID	Test scenario	Input	Expected result	Actual result	Status
TC-02	Analyst anchors an alert	raiseAlert(hash, wallet, 82, mask)	AlertRaised event emitted	Event emitted, alert stored	Pass
TC-07	Review workflow	Open → Reviewed → Escalated	Two AlertStatusChanged events	Both events emitted	Pass
TC-04	Negative: non-analyst raises alert	Call from an unapproved account	Revert	Reverted (not analyst)	Pass
TC-05	Negative: same tx reported twice	raiseAlert with an existing hash	Revert AlreadyReported	Reverted	Pass

15 / 15

Hardhat tests passed (8 negative)
+ 8 anchor tests, 54 Python tests

683,007 gas

Per raiseAlert on Sepolia
(deployment: 6,426,712 gas)

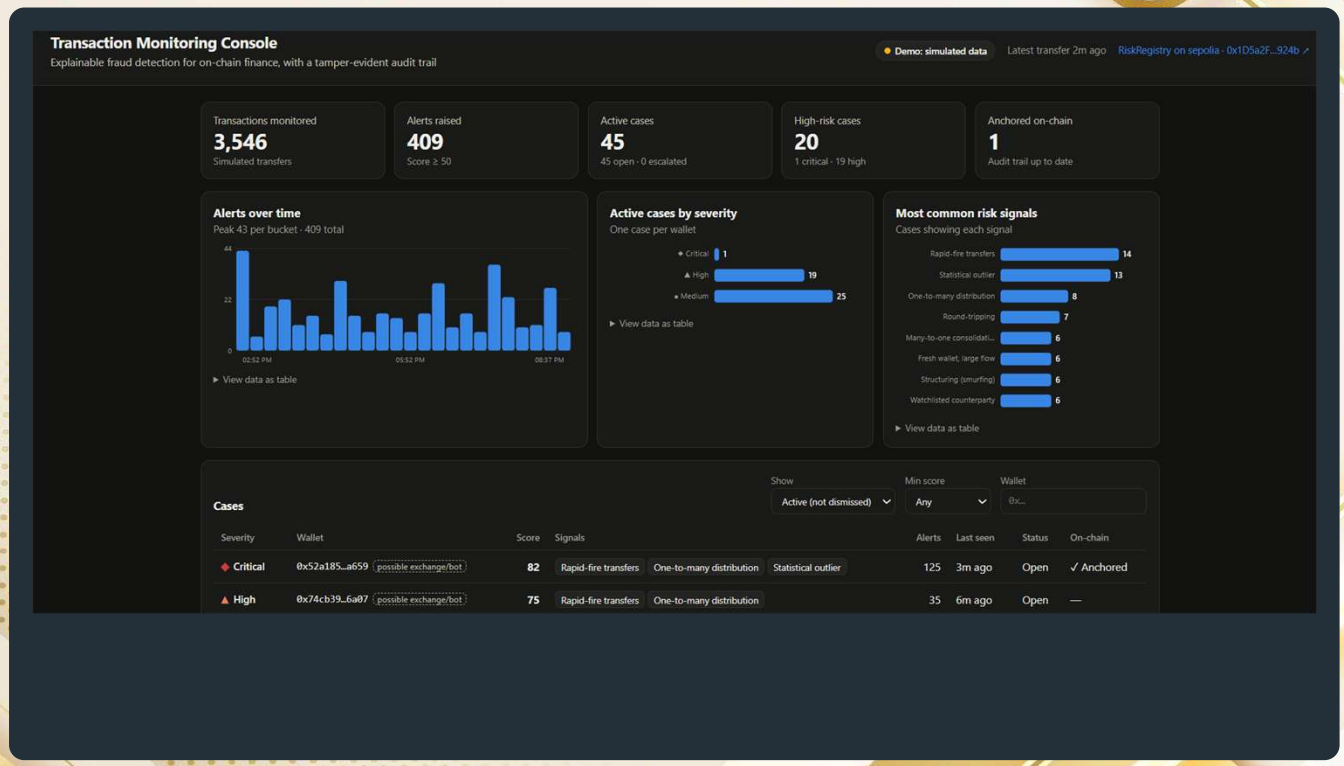
0x1D5a…924b

Deployed contract address
(Sepolia, block 12053731)

Detector on simulated fraud (seed 7, 3,546 transfers): 42/42 injected scenarios detected. Precision 0.996 with an exchange allowlist (1 false alert), 0.606 without it (161 false alerts). These figures come from simulated data and are not a claim of real-world accuracy.

Video Walkthrough

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Watch the demo video

Scan to watch

Video link: [https://drive.google.com/file/d/1XM5nRSkE7Bwcg5PfBr3fC0w2XQcKWQ9K/view?usp=sharin]

Objectives & Scope

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1. Ingest live Ethereum mainnet transfers through JSON-RPC into a local database (one live run added 2,365 transfers).
2. Detect seven AML patterns plus ML outliers and explain every alert (42 of 42 injected scenarios found on simulated data).
3. Deploy and test a role-based RiskRegistry contract on Sepolia (15 test cases, 8 of them negative).
4. Give analysts a dashboard to review cases and anchor alerts on-chain (alert #0 anchored on Sepolia).

In Scope

- Live ingestion of ETH transfers
- Rules + Isolation Forest, explainable score
- Case dashboard and review workflow
- On-chain anchoring on Sepolia

Out of Scope

- Mainnet deployment or real funds
- Blocking or freezing transactions
- ERC-20 and internal transfers (future extension)
- KYC / identity data

Literature Review

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Sl.	Author(s) & Year	Paper title / Source	Approach / Technique	Limitation / Gap
1	Weber et al., 2019	Anti-Money Laundering in Bitcoin: Experimenting with Graph Convolutional Networks for Financial Forensics – KDD Anomaly Detection in Finance Workshop	Graph convolutional network on a labelled Bitcoin transaction graph (Elliptic dataset)	Offline experiment on one labelled dataset; no live alerting, analyst workflow or audit trail
2	Chen et al., 2020	Phishing Scam Detection on Ethereum: Towards Financial Security for Blockchain Ecosystem – IJCAI 2020	Graph-based cascade features with a LightGBM dual-sampling ensemble	Supervised and needs labelled phishing accounts; classifies accounts but does not give explained, anchored alerts
3	Hassan et al., 2022	Anomaly Detection in Blockchain Networks: A Comprehensive Survey – arXiv:2112.06089	Survey of anomaly-detection models, metrics and requirements by blockchain layer	A survey: maps the field and open challenges but provides no working monitoring layer
4	Luo et al., 2023	AI-powered Fraud Detection in Decentralized Finance: A Project Life Cycle Perspective – arXiv:2308.15992	Taxonomy of DeFi fraud with statistical, ML and graph detection methods	Notes that early-stage detection is immature (little data, scarce ground truth); a review, not a tool
5	Liu, Ting & Zhou, 2008	Isolation Forest – IEEE ICDM 2008	Unsupervised anomaly detection by random isolation trees	General method; needs domain features and rules to be useful for AML and to be explained

Summary: existing work detects well offline, but none combines real-time monitoring, explainable scores and a tamper-evident review trail.

Conclusion & Future Scope

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Conclusion

- Built a working monitoring layer: live mainnet ingestion, eight rules plus Isolation Forest, and an explained 0–100 risk score per wallet.
- Deployed RiskRegistry on Sepolia and anchored a real alert (score 82, reasons: high velocity, fan-out, ML outlier) with its tx hash and gas shown on the dashboard.
- All 15 contract tests, 8 anchor tests and 54 Python tests pass; every objective on slide 4 was demonstrated.
- Live mode shows exchanges can raise false alarms, which is why allowlisting and human review are part of the design.

Future Scope

- Monitor ERC-20 tokens and internal transfers
- Add graph features and labelled data for supervised learning
- Batch anchoring and cheaper storage to cut gas
- Multi-signature analyst approval and role audit
- Deploy to an L2 and add alert webhooks for compliance tools

Key outcome: suspicious on-chain activity is detected in real time, explained in plain terms, and recorded in a tamper-evident audit trail.

References

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